

Database Functions (Self-Learning)

Q. What is a User-Defined Function (UDF) in SQL Explain how functions differ from Stored Procedures regarding parameters, return values, and execution contexts.

[DEF] Definition to Remember

A User-Defined Function (UDF) is a stored program that encapsulates SQL and procedural logic to perform a specific computation and must return a single value (scalar or table). Unlike procedures, functions are embedded directly inside SQL statements like SELECT, making them ideal for read-only calculations and data transformation.

1. Creating and Calling Functions

Creation: Must declare the return type explicitly.

```
CREATE
RETURN
AS
BEGIN
    RE
END;
```

Calling: Functions return a value, so they are embedded inside SQL -- NOT called via CALL.

```
SELECT
```

2. Parameters and Return Values

- * Parameters: Functions only accept IN parameters (input only -- no OUT or INOUT).

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3. Procedures vs Functions

Feature	Stored Procedure	User-Defined Function
Return Value	Optional; can return via OUT params	**Mandatory** -- must return a value
Called Using	`EXEC` or `CALL`	Embedded in SQL (`SELECT`, `WHERE`)
DML Operations	Can INSERT, UPDATE, DELETE	**Read-only** -- cannot modify data
Transactions	Can use COMMIT/ROLLBACK	Cannot manage transactions
Parameters	IN, OUT, INOUT	IN only
Best For	Workflows, data modification	Calculations, formatting, read-only lo

[MUST-WRITE] Must-Write Points (for 10 marks)

1. UDF = User-Defined Function -- extends SQL with custom calculations.
2. Created with CREATE FUNCTION; MUST declare return type and use RETURN.
3. Functions accept only IN parameters (no OUT or INOUT).
4. Functions are embedded directly inside SQL statements (not called with CALL).
5. Functions are read-only -- they CANNOT perform INSERT, UPDATE, or DELETE.
6. Procedures are optional return; Functions have mandatory return.
7. Scalar function -> single value; Table-valued function -> returns a table.

[RECALL] Quick Recall

UDF -> MUST return value (RETURNS + RETURN) -> IN params only -> Embedded in SELECT/WHERE -> Read-only (no DML) -> vs
Procedure: optional return, CALL, DML allowed